- Final Project
  - Final Project Proposal
    - 1. Abstract
    - 2. Motivation
    - 3. System Block Diagram
    - 4. Design Sketches
    - 5. Software Requirements Specification (SRS)
    - 6. Hardware Requirements Specification (HRS)
    - 7. Bill of Materials (BOM)
    - 8. Final Demo Goals
    - 9. Sprint Planning
  - Sprint Review #1
    - Last week's progress
    - Current state of project
    - Next week's plan
  - Sprint Review #2
    - Last week's progress
    - Current state of project
    - Next week's plan
  - MVP Demo
  - Final Report
    - 1. Video
    - 2. Images
    - 3. Results
      - 3.1 Software Requirements Specification (SRS) Results
      - 3.2 Hardware Requirements Specification (HRS) Results
    - 4. Conclusion
  - References

Review the assignment due date

Final Project

Team Number: 03

Team Name: Synth Specialists

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GitHub Repository URL: https://github.com/upenn-embedded/final-project-s26-t03.git

GitHub Pages Website URL: [for final submission]*

Final Project Proposal

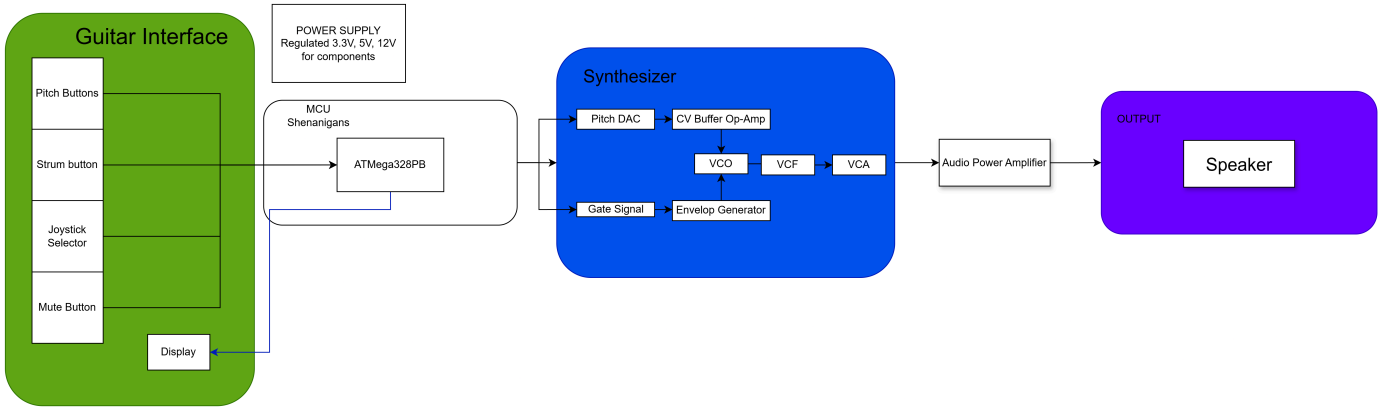
1. Abstract

Our project repurposes an old toy guitar controller into an embedded music-learning device. The controller's buttons will be programmable and mapped to individual notes or chords, while the strum bar will trigger sound generation. The sound will be produced by a synthesizer that we will design and build, and features like a mute button, whammy stick, and small display will make the device more expressive and customizable.

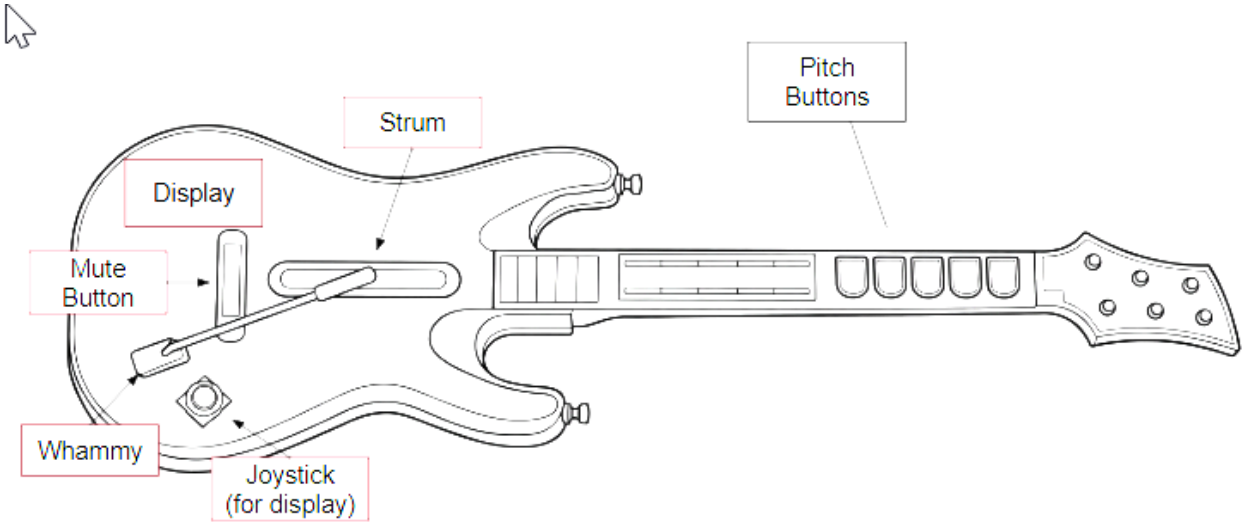
2. Motivation

Our motivation for choosing this project is that it combines hardware reuse with a meaningful educational application. Rather than leaving an obsolete gaming peripheral unused, we saw an opportunity to redesign it into a low-cost and engaging platform for teaching fundamental music concepts such as notes, chords, and pitch variation. The main problem we are trying to solve is that beginner music tools are often either too limited, too expensive, or not very interactive. By reusing an existing guitar controller and pairing it with a custom-built synthesizer, we can create a more hands-on and accessible way for users to learn basic musical ideas. This project is interesting because it combines embedded systems, digital sound generation, and interface design, while also showing how outdated electronics can be transformed into practical and creative learning tools.

3. System Block Diagram



4. Design Sketches



5. Software Requirements Specification (SRS)

5.1 Definitions, Abbreviations

MCU : ATmega328PB microcontroller

DAC : Digital-to-Analog Converter

VCO : Voltage-Controlled Oscillator

VCF : Voltage-Controlled Filter

VCA : Voltage-Controlled Amplifier

CV : Control Voltage

UI : User Interface

5.2 Functionality

ID	Description
SRS-01	The system shall read the guitar pitch buttons and determine the selected note or chord when a button is pressed.
SRS-02	When the user actuates the strum bar, the MCU shall generate and output the correspondingpitch control signal and gate signal.
SRS-03	The system shall read the whammy input through the ADC and update pitch bend during active note playback.
SRS-04	The system shall support a mute function that disables note output when the mute button being pressed.
SRS-05	The firmware shall continuously monitor all required inputs during operation, including pitch buttons, strum bar, mute button, joystick, and whammy control, without causing the system to miss valid note-trigger events during normal use
SRS-06	The display and joystick-based UI shall allow the user to change at least one synth setting (such as waveform, octave, or mode), and the selected setting shall be updated on the display.

6. Hardware Requirements Specification (HRS)

6.1 Definitions, Abbreviations

SPI : Serial Peripheral Interface

I2C : Inter-Integrated Circuit

ADC : Analog-to-Digital Converter

PWM : Pulse-Width Modulation

6.2 Functionality

ID	Description
HRS-01	The system shall use an ATmega328PB-based control system to interface with all required user inputs, including the pitch buttons, strum input, mute button, joystick, and whammy control
HRS-02	The system shall include a DAC-based pitch CV generation stage capable of producing a stable control signal for the synthesizer voice
HRS-03	The synthesizer hardware shall include, at minimum, a VCO, VCF, and VCA signal chain capable of producing an audible musical output in response to MCU-generated control signals
HRS-04	The system shall include an audio output stage consisting of a buffer and power amplifier capable of driving a speaker or external audio output
HRS-05	The system shall include a display module for showing menu or mode information and shall communicate with the MCU using a supported serial protocol such as I2C .
HRS-06	The system shall provide regulated power rails as required by the design, including support forlogic-level supply and analog/audio supply, so that all components operate within their rated voltage ranges.

7. Bill of Materials (BOM)

https://docs.google.com/spreadsheets/d/1yVx339OijMxmClf8FyQ_4eZAxMqQx_l-zpqn3cilw3A/edit?usp=sharing

8. Final Demo Goals

We will first demonstrate the functionality of the joystick selector and display by setting the buttons to specific chords. We will then play a song and showcase our synthesizer pitch bending with the whammy stick.

9. Sprint Planning

Milestone	Functionality Achieved	Distribution of Work
Sprint #1	Get Guitar and MCU communicating	We'll figure it out - but work together
Sprint #2	Get MCU and Synthesizer Working	~~
MVP Demo	Make Synth and Speaker work	~~
Final Demo	Have the whole product ready	~~

This is the end of the Project Proposal section. The remaining sections will be filled out based on the milestone schedule.

Sprint Review #1

Last week's progress

Current state of project

Next week's plan

Sprint Review #2

Last week's progress

Current state of project

Next week's plan

MVP Demo

Final Report

Don't forget to make the GitHub pages public website! If you’ve never made a GitHub pages website before, you can follow this webpage (though, substitute your final project repository for the GitHub username one in the quickstart guide): https://docs.github.com/en/pages/quickstart

1. Video

2. Images

3. Results

3.1 Software Requirements Specification (SRS) Results

ID	Description	Validation Outcome
SRS-01	The IMU 3-axis acceleration will be measured with 16-bit depth every 100 milliseconds +/-10 milliseconds.	Confirmed, logged output from the MCU is saved to "validation" folder in GitHub repository.

3.2 Hardware Requirements Specification (HRS) Results

ID	Description	Validation Outcome
HRS-01	A distance sensor shall be used for obstacle detection. The sensor shall detect obstacles at a maximum distance of at least 10 cm.	Confirmed, sensed obstacles up to 15cm. Video in "validation" folder, shows tape measure and logged output to terminal.

4. Conclusion

References